Questions on Algorithm

Q1. The worst-case running times of Insertion sort, Merge sortand Quick sort, respectively, are:

(a) $\theta(n\log n)$, $\theta(n\log n)$ and $\theta(n^2)$

(b) $\theta(n^2)$, $\theta(n^2)$ and $\theta(n\log n)$

(c) $\theta(n^2)$, $\theta(n\log n)$ and $\theta(n\log n)$

(d) $\theta(n^2)$, $\theta(n\log n)$ and $\theta(n^2)$

Q2. Which of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

(a) $O(\log n)$ (b) $O(n)$

(c) $O(n \log n)$ (d) $O(n^2)$

Q3. What is the number of swaps required to sort n elements usingselection sort, in the worst case?

(a) $\theta(n)$ (b) $\theta(n\log n)$

(c) $\theta(n^2)$ (d) $\theta(n^2 \log n)$

Q4. A sorting technique is called stable if

(a) It takes O (n logn) time

(b) It maintains the relative order of occurrence of non-distinctelements

(c) It uses divide and conquer paradigm

(d) It takes O(n) space

Q5. The number of swapping needed to sort the numbers 8, 22, 7,9, 31, 5, 13 in ascending order, using bubble sort is

(a) 11 (b) 12

(c) 13 (d) 10

Q6. How many comparisons are needed to sort an array of length 5 if a straight selection sort is used and array is already in the opposite order?

(a) 1 (b) 5

(c) 10 (d) 20

Q7. Selection sort, Quick sort is a stable sorting method

(a) True, True (b) False, False
(c) True, False (d) False, True

Q8. Given the array: 12, 40, 3, 2 and intermediate states of thearray while sorting

Stage (i) 12, 3, 40, 2
Stage (ii) 12, 3, 2, 40
Stage (iii) 3, 12, 2, 40
Stage (iv) 3, 2, 12, 40
Stage (v) 2, 3, 12, 40

The array has been sorted using

(a) Insertion sort
(b) Selection sort
(c) Quick sort
(d) Bubble sort

Q9. If the array A contains the items 10, 4, 7, 23, 67, 12 and 5 in that order, what will be the resultant array A after third pass of insertion sort?

(a) 67, 12, 10, 5, 4, 7, 23
(b) 4, 7, 10, 23, 67, 12, 5
(c) 4, 5, 7, 67, 10, 12, 23
(d) 10, 7, 4, 67, 23, 12, 5

Q10. Which one of the following is the recurrence equation for the worst case time complexity of the Quick sort algorithm for sorting n (≥ 2) numbers? In the recurrence equations given in the options below, c is a constant.

(a) $T(n) = 2T(n/2) + cn$
(b) $T(n) = T(n - 1) + T(0) + cn$
(c) $T(n) = 2T(n - 2) + cn$
(d) $T(n) = T(n/2) + cn$

Q11. Let P be a Quicksort Program to sort numbers in ascending order using the first element as pivot. Let t1 and t2 be the number of comparisons made by P for the inputs {1, 2, 3, 4, 5} and {4, 1, 5, 3, 2} respectively. Which one of the following holds?

(a) $t1 = 5$ (b) $t1 < t2$
(c) $t1 > t2$ (d) $t1 = t2$

Q12. What will be the output after second pass if we apply merge sort algorithm on following data:

20, 47, 15, 8, 9, 4, 40, 30, 12, 17

Ans:

1) D
2) B
3) A
4) B
5) D
6) C
7) B
8) D
9) B
10) B
11) C

Questions on Hashing

Q1. Consider a hash table of size seven, with starting index zero, and a hash function (3x + 4) mod 7. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table. (a) 8, _, _, _, _, _, 10
(b) 1, 8, 10, _, _, _, 3
(c) 1, _, _, _, _, _,3
(d) 1, 10, 8, _, _, _, 3

Q2. The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function h(k) = k mod 10 and linear probing. What is the resultant hash table?

(a)

0	
1	
2	2
3	23
4	
5	15
6	
7	
8	18
9	

(b)

0	
1	
2	12
3	13
4	
5	5
6	
7	
8	18
9	

(b)

0	
1	
2	12
3	13
4	2
5	3
6	23
7	5
8	18
9	15

(d)

0	
1	
2	12, 2
3	13, 3, 23
4	
5	5, 15
6	
7	
8	18
9	

Q3. A hash table of length 10 uses open addressing with hash function h(k)=k mod 10, and linear probing. After inserting 6 values into an empty hash table, the table isas shown below:

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	
10	

Which one of the following choices gives a possible order in which the key valuescould have been inserted in the table?

(a) 46, 42, 34, 52, 23, 33 (b) 34, 42, 23, 52, 33, 46

(c) 46, 34, 42, 23, 52, 33 (d) 42, 46, 33, 23, 34, 52

Q4. Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first 3 insertions?

(a) (97 × 97 × 97)/100^3 (b) (99 × 98 × 97)/100^3

(c) (97 × 96 × 95)/100^3 (d) (97 × 96 × 95)/(3! × 100^3)

Q5. Which one of the following hash functions on integers will distribute keys mostuniformly over 10 buckets numbered 0 to 9 for i ranging from 0 to 2020?

(a) h(i) =i^2 mod 10

(b) h(i) =i^3 mod 10

(c) h(i) = (11 ∗ i^2) mod 10

(d) h(i) = (12 ∗ i) mod 10

Q6. A hash function h defined h(key)=key mod 7, with linear probing, is used to insert the keys 44, 45, 79, 55, 91, 18, 63 into a table indexed from 0 to 6. What will be the location of key 18?

(a) 3

(b) 4

(c) 5

(d) 6

Q7. Insert the characters of the string K R P C S N Y T J M into a hash table of size 10. Use the hash function h(x) = (ord(x) – ord("a") + 1) mod10 If linear probing isused to resolve collisions, then the following insertion causes collision

(a) Y

(b) C

(c) M

(d) P

Q8. Consider a 13 element hash table for which f(key)=key mod 13 is used with integer keys. Assuming linear probing is used for collision resolution, at which location would the key 103 be inserted, if the keys 661, 182, 24 and 103 are insertedin that order?

(a) 0

(b) 1

(c) 11

(d) 12

Ans:

1. B
2. C
3. C

4. A
5. B
6. C
7. C
8. B

Questions on Python

Q1. Which one of these is floor division?

a) /
b) \\
c) //
d) %

Q2. What is the output of this expression, 5*1**4?

a) 4
b) 5
c) 20
d) 625

Q3. Which of the following is used to define a block of code in Pythonlanguage?

a) Brackets
b) Key
c) Indentation
d) All of the mentioned

Q4. What will be the output of the following Python code?i = 1

```
while True:
    if i%3 == 0:
        break
    print(i)
    i +  =  1
```

a) 1 2 3
b) 1 2
c) 1
d) Syntax error

Q5. What are the values of the following Python expressions?

```
3**(2**3)
3**2**3
(3**2)**3
```

a) 6561, 729, 729
b) 6561, 6561, 729
c) 6561, 729, 6561
d) 6561, 6561, 6561

Q6. What will be the output of the following Python function?min(max(False,0,-3,-4), 2,7)

a) -4 b) 0

c) 7 d) False

Q7. What will be the output of the following Python code snippet? for a in [5, 7, 9,3][::-1]:

```
    print (a)
```

a) 3 9 7 5 b) Syntax error

c) 5 7 9 3 d) None of these

Q8. Which of these is not a valid Python variable name?

a) firstVariable b) first_variable

c) _variable d) 1st_var

Q9. Find output generated by the following code:p=6/4

```
q=p*2
r=p+q+1
p=p*q+r
q=p*q+r
r=p*q+r
print(r,p,q)
```

a) 350.0, 10.0, 35.0

b) 360.5, 10.0, 35.5

c) 350.5 10.0, 35.0

d) 350.0, 10.0, 35.5

Q10. Find output generated by the following code:

```
3+5*7//4+2**3*2
```

a) 28 b) 26

c) 28.25 d) 27

Q11. What will be the output of the following Python code?def first(x):

```
    return x ** 2
def first(x):
    return x ** 3
print(first(3))
```

Q12. What will be the output of the following Python code?def fn():

```
        global var1
        var1=20
        var1=var1*3
        print(var1)
var1=15
fn()
print(var1)
```

a) 45
15

b) 60
15

c) 45
45

d) 60
60

Q13. What will be the output of the following Python code?def first():

```
        second()
        print('first')
def second():
         third()
        print('second')
def third():
        print('third')
        first()
```

Q14. What will be the output of the following Python code?def recursion(n):

```
    if n == 1:
        return 1
    return recursion(n - 1) + nN = 5
print(recursion(N))
```

Q15. What will be the output of the following Python code?q=10

```
        p=5 p*=q//3
        q+=p+q**2
        print(p,q)
```

Q16. What will be the output of the following Python code?

```
        Str="gate smashers"
        s=Str[-5:]
        s*2
```

Q17. What will be the output of the following Python code?x="first"

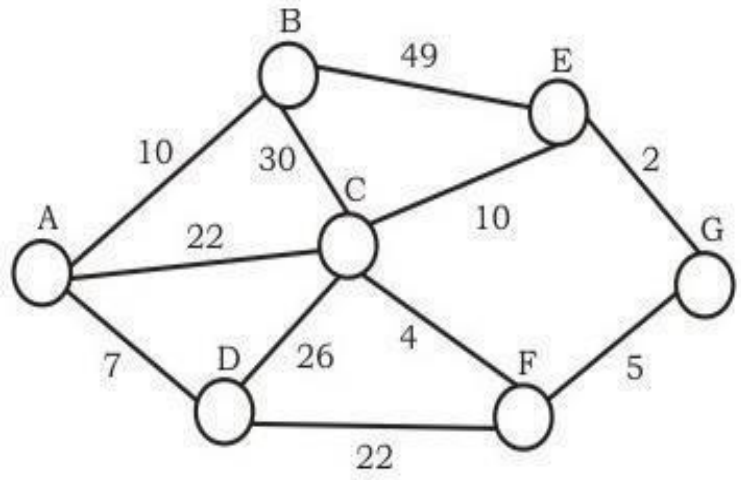
```
y="second"
c=0
while c<len(x):
    print(x[c],y[c])
    c=c+1
```

Ans:

1. C
2. B
3. C
4. D
5. B
6. D
7. A
8. D
9. B
10. D
11.-
12. D

Questions on Spanning Tree, BFS & DFS

Q1. Consider the undirected graph below:



Using Prim's algorithm to construct a minimum spanning tree starting with node A, which one of the following sequences of edges represents a possible order in which the edges would

be added to construct the minimum spanning tree?

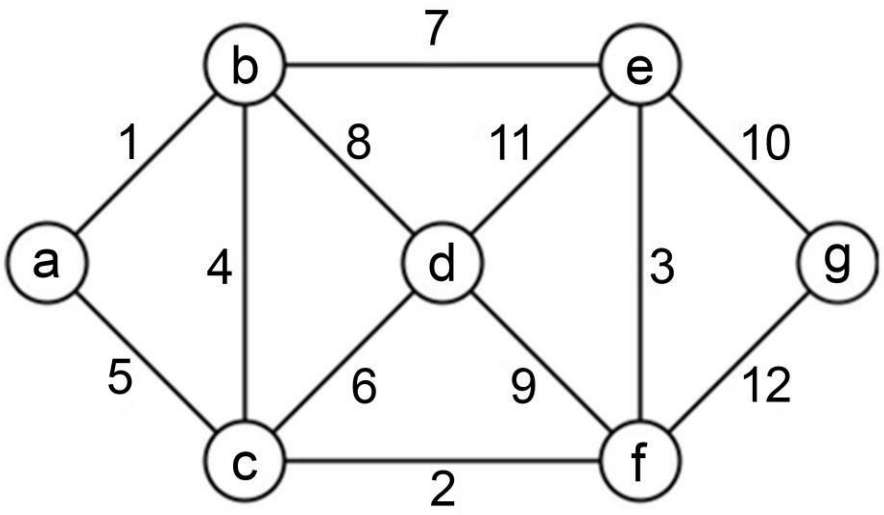
(a) (E, G), (C, F), (F, G), (A, D), (A, B), (A, C)

(b) (A, D), (A, B), (A, C), (C, F), (G, E), (F, G)

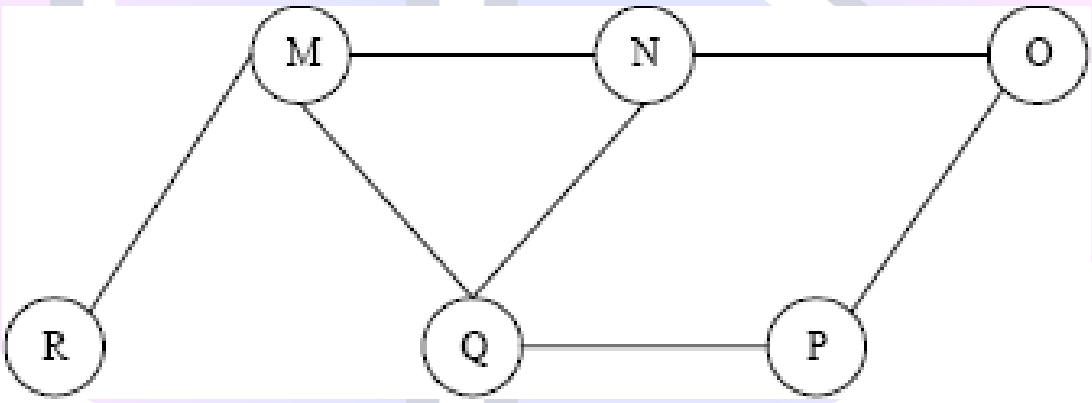
(c) (A, B), (A, D), (D, F), (F, G), (G, E), (F, C)

(d) (A, D), (A, B), (D, F), (F, C), (F, G), (G, E)

Q2. Using Prim's Algorithm, find the cost of minimum spanning tree (MST) ofthe given graph



Q3. The Breadth First Search algorithm has been implemented using the queue data structure. One possible order of visiting the nodes of the following graph is:



(a) MNOPQR

(b) NQMPOR

(c) QMNPRO

(d) QMNPOR

Q4. Consider the following graph

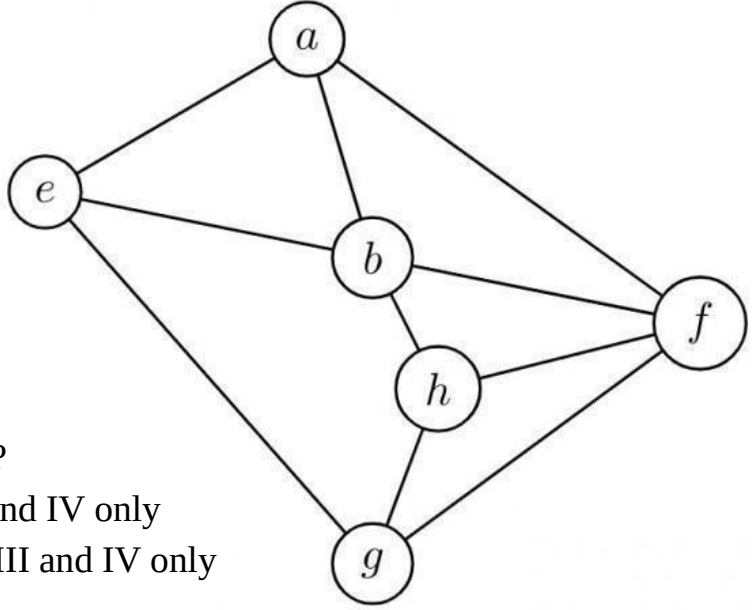
Among the following sequences:

i) a b e g h f

ii) a b f e h g

iii) a b f h g e

iv) a f g h b e



Which are depth first traversals of the above graph?

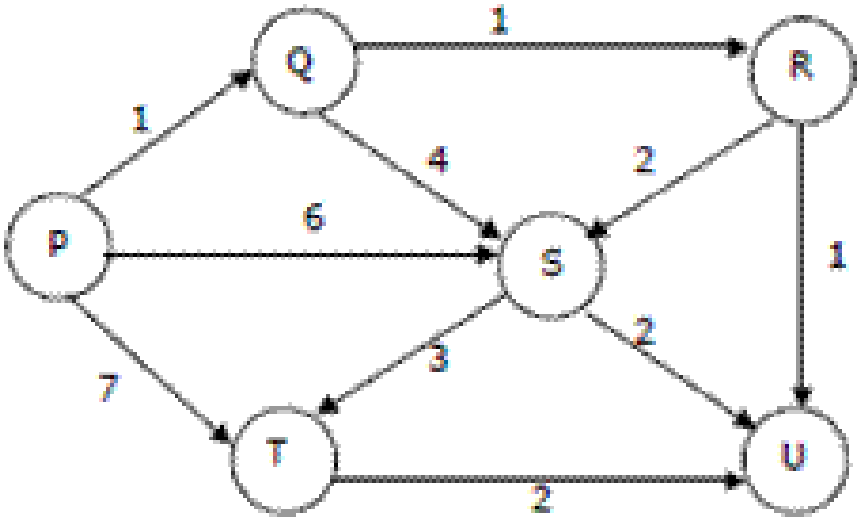
(a) I, II and IV only

(b) I and IV only

(c) II, III and IV only

(d) I, III and IV only

Q5. Suppose we run Dijkstra's single source shortest-path algorithm on the following edge weighted directed graph with vertex P as thesource. In what order do the nodes get included into the set of vertices for which the shortest path distances are finalized?



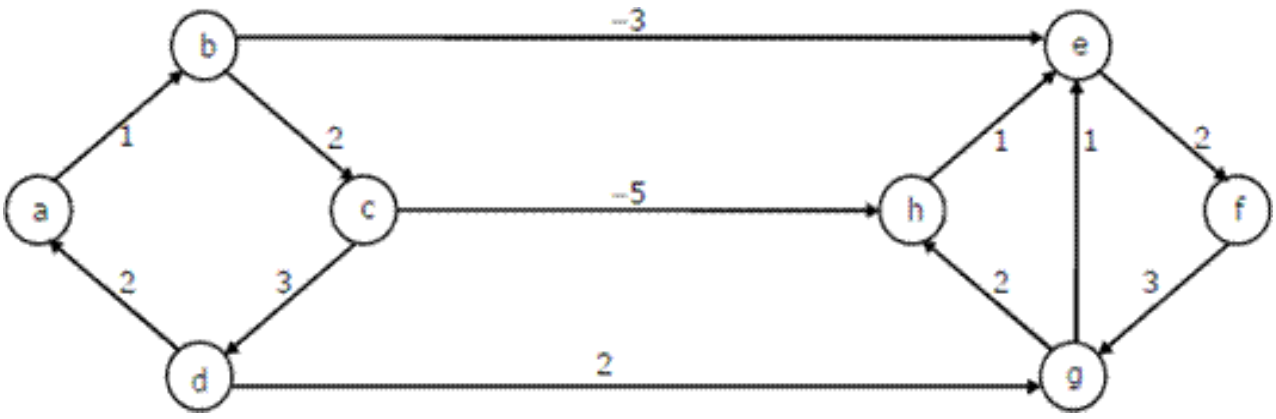
(a) P, Q, R, S, T, U

(b) P, Q, R, U, S, T

(c) P, Q, R, U, T, S

(d) P, Q, T, R, U, S

Q6. Dijkstra's single source shortest path algorithm when run from vertexa in the below graph,computes the correct shortest path distance to



(a) only vertex a

(b) only vertices a, e, f, g, h

(c) only vertices a, b, c, d

(d) all the vertices

Q7. Let G be an undirected graph. Consider a depth-first traversal of G, and let T be the resulting depth-first search tree. Let u be a vertex in G and let v be the first new (unvisited) vertex visited after visiting u in the traversal. Which of the following statements is always true?

(a) {u,v} must be an edge in G, and u is a descendant of v in T

(b) {u,v} must be an edge in G, and v is a descendant of u in T

(c) If {u,v} is not an edge in G then u is a leaf in T

(d) If {u,v} is not an edge in G then u and v must have the same parent inT

Ans:

1) D
2) --
3) C
4) D
5) B
6) D
7) C